

NO CYCLIC SKEW FRAME STARTERS OF TYPES 6^8 AND 6^9 : AN EVEN-ORDER SUM-OF-SQUARES OBSTRUCTION

ABSTRACT. Let H be the subgroup of order h of $\mathbb{Z}_g$, where $g = hu$, and put $T = \mathbb{Z}_g \setminus H$ and $Q = \sum_{z \in T} z^2$. We prove that if a cyclic skew frame starter of type h^u exists in $\mathbb{Z}_g \setminus H$ then $Q \equiv 0 \pmod{g}$, *with no hypothesis on the parity of g* ; for odd g this is a theorem of Stinson, whose proof halves and therefore needs 2 invertible. Since $Q \equiv 40 \pmod{48}$ for the type 6^8 and $Q \equiv 36 \pmod{54}$ for the type 6^9 , there is no cyclic skew frame starter of either type. These are two of the three cells left open in Table 1 of Stinson's paper. The third, 4^{11} , has $Q \equiv 0 \pmod{44}$, so the criterion is silent there and that cell remains open.

1. THE OPEN CELLS

Let G be an additive abelian group of order g and $H \leq G$ a subgroup of order h with $g - h$ even; put $m = (g - h)/2$. Following Stinson [2, §1], a *frame starter* in $G \setminus H$ is a set

$$S = \{\{x_i, y_i\} : 1 \leq i \leq m\}$$

of m unordered pairs such that

$$\begin{aligned} (1) \quad & \{x_i, y_i : 1 \leq i \leq m\} = G \setminus H, \\ (2) \quad & \{\pm(x_i - y_i) : 1 \leq i \leq m\} = G \setminus H, \end{aligned}$$

both as multisets, i.e. every element of $G \setminus H$ occurs exactly once as an entry and exactly once as a signed difference. The *type* of S is $h^{g/h}$. The starter is *skew* if moreover

$$(3) \quad \{\pm(x_i + y_i) : 1 \leq i \leq m\} = G \setminus H.$$

Skew frame starters produce skew Room frames, and a skew frame starter is in particular strong. Throughout, $G = \mathbb{Z}_g$ is *cyclic*, so $H = \langle u \rangle$ with $u = g/h$ is the unique subgroup of order h , and we write

$$T = \mathbb{Z}_g \setminus H, \quad Q = \sum_{z \in T} z^2 \in \mathbb{Z}.$$

All congruences are modulo g unless said otherwise.

2020 *Mathematics Subject Classification.* 05B15, 05B30, 05B10.

Key words and phrases. skew frame starter, frame starter, Room frame, cyclic group, sum of squares, nonexistence.

Table 1 of [2] tabulates, for cyclic groups and $u > 1$, the existence of a skew frame starter of type h^u for $h \leq 8$, with a named authority for each decided cell. Exactly three of its cells are left open:

$$4^{11} (\mathbb{Z}_{44}), \quad 6^8 (\mathbb{Z}_{48}), \quad 6^9 (\mathbb{Z}_{54}).$$

Theorem 1. *There is no cyclic skew frame starter of type 6^8 and none of type 6^9 . Explicitly, no set of 21 pairs satisfies (1), (2), (3) in $\mathbb{Z}_{48} \setminus \langle 8 \rangle$, and no set of 24 pairs satisfies them in $\mathbb{Z}_{54} \setminus \langle 9 \rangle$.*

2. THE CRITERION, AT EVERY PARITY

Lemma 2 (parity). *Suppose $H \leq \mathbb{Z}_g$ has order h and $g - h$ is even. Then g and h have the same parity, and if g is even then $g/2 \in H$. Consequently T contains no self-inverse element, and*

$$T = \mathcal{H} \sqcup (-\mathcal{H}), \quad \mathcal{H} := \left\{ j : 1 \leq j \leq \lfloor \frac{g-1}{2} \rfloor, j \notin H \right\}, \quad |\mathcal{H}| = m.$$

Proof. $g - h$ even is exactly “ $g \equiv h \pmod{2}$ ”. If g is even then so is h , and $g/2 = u \cdot (h/2)$ is a multiple of u , hence lies in $H = \langle u \rangle$. The self-inverse elements of $\mathbb{Z}_g$ are 0 and, when g is even, $g/2$; both lie in H . So $z \mapsto -z$ is a fixed-point-free involution of T , and it pairs each $j \in \mathcal{H}$ with $g - j \notin \mathcal{H}$, which gives the decomposition and $|\mathcal{H}| = |T|/2 = m$. $\square$

Write $\Theta = \sum_{j \in \mathcal{H}} j^2 \in \mathbb{Z}$. Lemma 2 gives an identity in which no starter appears:

$$(4) \quad Q = \sum_{j \in \mathcal{H}} (j^2 + (g - j)^2) \equiv 2\Theta \pmod{g}.$$

Theorem 3. *If a cyclic skew frame starter of type h^u exists in $\mathbb{Z}_g \setminus H$, then $Q \equiv 0 \pmod{g}$. No hypothesis on the parity of g is used.*

Proof. Let $S = \{\{x_i, y_i\}\}$ be such a starter and put $d_i = x_i - y_i$, $s_i = x_i + y_i$ in $\mathbb{Z}_g$; choose integer representatives once and for all, all congruences being mod g . By (1), $\sum_i (x_i^2 + y_i^2) \equiv Q$. By (2) the $2m$ residues $\pm d_i$ are exactly T , so by Lemma 2 the map sending i to the unique $j_i \in \mathcal{H}$ with $d_i \equiv \pm j_i$ is a bijection $\{1, \dots, m\} \rightarrow \mathcal{H}$; since $(\pm j_i)^2 = j_i^2$,

$$\sum_i d_i^2 \equiv \sum_{j \in \mathcal{H}} j^2 = \Theta.$$

By (3) the same argument applied to the sums gives $\sum_i s_i^2 \equiv \Theta$. Summing the integer identity $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$ over i yields

$$\sum_i s_i^2 + \sum_i d_i^2 \equiv 2Q,$$

hence $2\Theta \equiv 2Q$. Substituting (4) in the form $Q \equiv 2\Theta$ gives $Q \equiv 2Q$, that is $Q \equiv 0 \pmod{g}$. No step divides by 2, which is why g need not be odd. $\square$

Lemma 4 (elementary form). *Put $N = (g-1)(2g-1) - u(h-1)(2h-1)$. Then $Q = gN/6$, and $Q \equiv 0 \pmod{g}$ holds if and only if $6 \mid N$, which in turn holds if and only if both*

- (i) *if h is even then u is odd; and*
- (ii) *if $3 \mid g$ then $3 \mid h$ and $u \equiv 1 \pmod{3}$.*

Proof. $\sum_{z=0}^{g-1} z^2 = g(g-1)(2g-1)/6$ and $\sum_{z \in H} z^2 = \sum_{k=0}^{h-1} (uk)^2 = u^2 h(h-1)(2h-1)/6 = gu(h-1)(2h-1)/6$, since $u^2 h = ug$. Subtracting gives $Q = gN/6$. Write $N = 6q + r$ with $0 \leq r < 6$; then $Q = gq + gr/6$, and $gr/6$ is an integer because Q and gq are, so $Q \equiv gr/6 \pmod{g}$ with $0 \leq gr/6 < g$. Hence $Q \equiv 0$ iff $r = 0$.

Modulo 2: $N \equiv (g+1) + u(h+1)$. If h is even then $g = hu$ is even, so $N \equiv 1 + u$ and $2 \mid N$ iff u is odd. If h is odd then, by Lemma 2, g is odd and hence u is odd, so $g+1$ and $h+1$ are both even and $2 \mid N$ automatically.

Modulo 3: if $3 \nmid g$ then $3 \nmid h$ and $3 \nmid u$, and running through the four possibilities for $(g \pmod{3}, h \pmod{3})$ one finds $(g-1)(2g-1) \equiv 0$ and $u(h-1)(2h-1) \equiv 0$ in each, so $3 \mid N$. If $3 \mid g$ then $(g-1)(2g-1) \equiv 1$, and $3 \mid N$ iff $u(h-1)(2h-1) \equiv 1$; this forces $3 \nmid h-1$ and $3 \nmid 2h-1$, i.e. $3 \mid h$, in which case $(h-1)(2h-1) \equiv 1$ and the condition becomes $u \equiv 1 \pmod{3}$. $\square$

Proof of Theorem 1. For type 6^9 we have $g = 54$, $h = 6$ and $u = 9$, so that $H = \{0, 9, 18, 27, 36, 45\}$ and $m = 24$, and

$$N = 53 \cdot 107 - 9 \cdot 5 \cdot 11 = 5671 - 495 = 5176 = 6 \cdot 862 + 4,$$

so $6 \nmid N$ and $Q = 54 \cdot 5176/6 = 46,584 = 54 \cdot 862 + 36$, i.e. $Q \equiv 36 \not\equiv 0 \pmod{54}$. In the elementary form, clause (i) holds ($u = 9$ is odd) and clause (ii) fails, because $3 \mid 54$ and $3 \mid 6$ but $u = 9 \equiv 0 \pmod{3}$.

For type 6^8 : $g = 48$, $h = 6$, $u = 8$, $H = \{0, 8, 16, 24, 32, 40\}$, $m = 21$, and

$$N = 47 \cdot 95 - 8 \cdot 5 \cdot 11 = 4465 - 440 = 4025,$$

which is odd, so $6 \nmid N$ and $Q = 48 \cdot 4025/6 = 32,200 = 48 \cdot 670 + 40$, i.e. $Q \equiv 40 \not\equiv 0 \pmod{48}$. Clause (i) fails: $h = 6$ is even and $u = 8$ is even.

In both cases Theorem 3 is contradicted, so no such starter exists. $\square$

Remark (the third open cell). For $g = 44$, $h = 4$, $u = 11$ one has $N = 43 \cdot 87 - 11 \cdot 3 \cdot 7 = 3741 - 231 = 3510 = 6 \cdot 585$, so $6 \mid N$ and $Q \equiv 0 \pmod{44}$: both clauses of Lemma 4 hold, and Theorem 3 is *provably silent* on the cell 4^{11} , which remains open.

Remark (scope). Only cyclic groups are treated. Nothing here bears on the noncyclic abelian groups of order 48 or 54, and the distinction is not a formality: [2] records types admitting no cyclic skew frame starter but admitting noncyclic ones. A skew Room *frame* of type 6^9 exists [1]; what Theorem 1 excludes is the starter construction in a cyclic group.

Remark (attribution). For g odd, Theorem 3 is Theorem 2.4 of [2], whose Lemma 2.3 evaluates $Q = g(2gh-1)(g-h)/(6h)$ and whose condition $(2gh-1)(g-h) \not\equiv 0 \pmod{6h}$ is the same congruence. That proof passes

through the patterned-starter theorem of [2] and sets $a_i = (x_i + y_i)/2$, so it needs 2 invertible; for even order cyclic groups [2] uses a different approach. The only new content in Theorem 3 is the removal of “ g odd”, obtained through Lemma 2 and (4). The sum-of-squares technique itself is credited in [2] to Constable and to Wallis and Mullin. Besides the two cells of Theorem 1, the criterion gives noncomputational proofs of nonexistence for seven of the nine types that [2] settles by exhaustive search; all seven of those have g even.

3. VERIFICATION

Both theorems can be redone at a desk: form $H = \langle g/h \rangle$, sum z^2 over $\mathbb{Z}_g \setminus H$, and compare with $gN/6$. The accompanying program `verify.py` does this in exact integer arithmetic using only the Python standard library. It checks Lemma 2 and the identity (4) cell by cell, recomputes Q two independent ways, verifies that the clause form of Lemma 4 agrees with $Q \equiv 0 \pmod{g}$, and reproduces the arithmetic of the two cells of Theorem 1 and the silence at 4^{11} . It runs these checks across the types tabulated in Table 1 of [2] and across further material not reproduced in this paper, and its transcript `verify.output.txt` reports 70 checks, all passing. Neither Theorem 1 nor Theorem 3 depends on a machine.

REFERENCES

- [1] K. Chen and L. Zhu, *On the existence of skew Room frames of type t^u* , *Ars Combin.* **43** (1996), 65–79.
- [2] D. R. Stinson, *Some new results on skew frame starters in cyclic groups*, *Discrete Math.* **346** (2023), no. 9, Art. 113476. doi:10.1016/j.disc.2023.113476; arXiv:2211.12367v1. Table 1 and the numbering of results cited here are those of the arXiv version; the journal version was not accessible to us and may number differently.